Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Welcome to the Course

- Welcome to Applied Unsupervised Learning in Python
  Video • 5 min
- MADSwPy Certificate Roadmap
  Reading • 5 min
- Course Syllabus
  Reading • 10 min
- **Additional Resources**
  Reading • 10 min
- Meet Your Fellow Learners
  Discussion Prompt • 15 min
- Help Us Learn About You
  Reading • 5 min

Lesson 1: Methods for Dimensionality Reduction

- Dimensionality Reduction: A Brief Introduction
  Video • 16 min
- Dimensionality Reduction with Feature Selection: Information Gain
  Video • 15 min
- Time to Practice: Dimensionality Reduction
  Graded Assignment • Grade: 100%
- Dimensionality Reduction with Feature Selection: Principal Component Analysis (PCA) Explained
  Video • 20 min
- Visualizing PCA Results: Foundations
  Video • 8 min
- Visualizing PCA Results: Biplots and Variance Plots
  Video • 17 min
- Ten Quick Tips for Effective Dimensionality Reduction
  Reading • 45 min
- Time to Practice: Principal Component Analysis (PCA)
  Graded Assignment • Grade: 100%
- Singular Value Decomposition (SVD)
  Video • 21 min
- Applications of SVD in Data Science
  Video • 13 min
- Time to Practice: Singular Value Decomposition (SVD)
  Graded Assignment • Grade: 100%
- Manifold Learning: Multidimensional Scaling (Part 1)
  Video • 17 min
- Manifold Learning: Multidimensional Scaling (Part 2)
  Video • 11 min
- Time to Practice: Manifold Learning (Multidimensional scaling: Parts 1 & 2)
  Graded Assignment • Grade: 100%
- Manifold Learning: t-Distributed Stochastic Neighbor Embedding (t-SNE)
  Video • 13 min
- Manifold Learning: Uniform Manifold Approximation and Projection (UMAP)
  Video • 6 min
- Time to Practice: Manifold Learning (t-SNE, and UMAP)
  Graded Assignment • Grade: 100%

Lesson 2: Density Estimation

- Density Estimation Part 1: Probability Density Functions
  Video • 12 min
- Density Estimation Part 1: Parametric vs. Non-Parametric Density Estimator
  Video • 13 min
- Time to Practice: Density Estimation Methods (Part 1)
  Graded Assignment • Grade: 100%
- Density Estimation Part 2: Local Density Estimators
  Video • 11 min
- Density Estimation Part 2: Kernel Density Estimators
  Video • 15 min
- Density Estimation Part 2: Evaluating Density Estimators
  Video • 4 min

Additional Resources

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To succeed in this course, it is helpful to have background knowledge of basic concepts in linear algebra, statistics, and machine learning. For a refresher, we recommend the following resources:

Linear Algebra

- Essence of Linear Algebra
  - Optional: Chapter 6 (Determinant) and Chapter 12 (Cramer's Rule)

Statistics

- Statistics with Python Using NumPy, Pandas, and SciPy
- Understanding Data: How to Describe Data

Machine Learning

- Introduction to Machine Learning

Go to next item ✓ Completed

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